MAT 534 HW11

Aditya Pahuja

Due 12/08

Problem 1. Let R be a unique factorization domain and $d \in R$ a nonzero element. Show that there are only finitely many distinct principal ideals containing d.

Solution. Let $u \cdot p_1^{e_1} \cdots p_k^{e_k}$ be a factorization of d where the p_i are irreducible, the e_k are positive integers, and u is a unit. Then $d \in (c)$ for some $c \in R$ if and only if $c \mid d$, which is true if and only if $c = u' \cdot p_1^{e_1'} \cdots p_k^{e_k'}$ for nonnegative integers $e_k' \leq e_k$ and unit u'. Two choices of c generate the same ideal if and only if the tuples $(e_1', \ldots, e_k')$ are the same (i.e. factorizations are the same up to units), and there are $(e_1+1)\cdots(e_k+1) < \infty$ choices for this tuple ($e_i + 1$ choices for $e_i' \in \{0, 1, \ldots, e_i\}$), hence only finitely many principal ideals containing d. $\square$

Problem 2. Let $A \in GL_n(F)$. Prove that there is a polynomial $P(x) \in F[x]$ of degree at most n such that $A^{-1} = P(A)$.

Solution. Let $Q(x) \in F[x]$ be the characteristic polynomial of A, which has degree n. By the Cayley-Hamilton theorem, $Q(A) = 0$, and since A is invertible, $Q(0) = \det A \neq 0$. Writing $Q(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$, define

$$P(x) = -\frac{Q(x) - a_0}{a_0 x}$$

so that

$$AP(A) = -\frac{1}{a_0}Q(A) + I = I \iff P(A) = A^{-1}. \qquad \square$$

Problem 3. Describe the field of fractions of the ring $\mathbb{C}[x,y]/(y^2 - x^3)$.

Solution. First, observe that the surjective homomorphism $\varphi\colon \mathbb{C}[x,y] \to \mathbb{C}[t^2,t^3]$ given by $\varphi(x) = t^2$ and $\varphi(y) = t^3$ has kernel $(y^2 - x^3)$. Clearly the kernel contains this ideal; suppose that $p(x,y) \in \ker\varphi$. In $\mathbb{C}[x,y]/(y^2-x^3)$, $y^2 = x^3$, so for any polynomial in $\mathbb{C}[x,y]$, we can find a polynomial in the same coset whose y-degree is reduced to zero or 1; in $\mathbb{C}[x,y]$ this corresponds to writing $p(x,y) = q(x,y)(y^2 - x^3) + r_1(x)y + r_0(x)$. Then

$$\varphi(p) = r_1(t^2)t^3 + r_0(t^2).$$

The degrees of the monomials in $r_1(t^2)t^3$ are all odd, while the degrees of the monomials in $r_0(t^2)$ are all even, so the only way that the polynomial on the right is zero is if r_1 and r_0 are the zero polynomial, implying $p(x,y) = q(x,y)(y^2 - x^3) \in (y^2 - x^3)$.

Therefore we want to describe the fraction field of $\mathbb{C}[t^2,t^3]$. Since $t = t^3/t^2 = \phi(y)/\phi(x)$ is in this fraction field, $\mathrm{Frac}(\mathbb{C}[t^2,t^3]) = \mathrm{Frac}(\mathbb{C}[t]) = \mathbb{C}(t)$. In terms of the original ring, the fraction field of $\mathbb{C}[x,y]/(y^2-x^3)$ is $\mathbb{C}(\overline{y}/\overline{x})$ where $\overline{x}, \overline{y} \in \mathbb{C}[x,y]/(y^2-x^3)$ are the projections of x and y to the quotient. $\square$

Problem 4. Let $T\colon \mathbb{R}^4 \to \mathbb{R}^4$ be a linear map such that the minimal polynomial of T is x^4+1. Determine the number of linear subspaces $W \subseteq \mathbb{R}^4$ such that $T(W) \subseteq W$.

Solution. Treat $\mathbb{R}^4$ as an $\mathbb{R}[x]$-module where the action of x is given by applying T. By the classification of finitely generated modules over PIDs, we have module isomorphism

$$\mathbb{R}^4 \cong \mathbb{R}[x]/(x^2 - x\sqrt{2} + 1) \oplus \mathbb{R}[x]/(x^2 + x\sqrt{2} + 1)$$

using that the annihilator of $\mathbb{R}^4$ is $x^4 + 1 = (x^2 - x\sqrt{2} + 1)(x^2 + x\sqrt{2} + 1)$ and matching the dimensions to choose the exponents of the quadratics.

Since T has no real eigenvalues ($x^4 + 1$ has no real roots), T does not fix any one-dimensional subspace.

Among two-dimensional subspaces, there are three cases: a two-dimensional subspace of the decomposed module is one of $\mathbb{R}[x]/(x^2 - x\sqrt{2} + 1)$, $\mathbb{R}[x]/(x^2 + x\sqrt{2} + 1)$, or a direct sum of a one-dimensional subspace of each factor. Since neither one-dimensional subspace is fixed by multiplication by x, a subspace of the third kind cannot be mapped into itself by T, while the other two subspaces of course are mapped to themselves via multiplication by x, giving two 2-dimensional subspaces that are fixed by T.

A 3-dimensional subspace corresponds to a direct sum of one of the factors of the decomposition with a one-dimensional subspace of the other factor, which cannot be fixed by multiplication by x because the one-dimensional subspace won't be mapped into itself.

There are also two trivial subspaces satisfying $T(W) \subseteq W$, namely $W = \{0\}$ and $W = \mathbb{R}^4$, for a total of $\boxed{4}$ such subspaces W. □

Problem 5. Show that a group of order 72 cannot be simple.

Solution. The number of Sylow 3-subgroups (of order 9) is 1 mod 3 and a divisor of 8, hence either 1 or 4. If there is only one then it must be normal, so suppose there are four. Then, if N is a normalizer of one of these Sylow 3-subgroups, $|G|$ does not divide $(G : N)! = 24$, which implies that G is not simple by HW02 #6(d). □

Problem 6. Find, up to isomorphism, all finite groups that have precisely three conjugacy classes.

Solution. Let G be a group of order $n \in \mathbb{N}$ with three conjugacy classes. One of these conjugacy classes is $\{1\}$; denote the other two by C_1 and C_2. We know that $|C_1|$ and $|C_2|$ are proper divisors of n and $1 + |C_1| + |C_2| = n$; this means that either $1 = |C_1| = |C_2|$ or one of $1, |C_1|, |C_2|$ must be larger than $n/3$.

The former case of course corresponds to $n = 3$, i.e. $G \cong \mathbb{Z}/3\mathbb{Z}$. In the latter case, the only possible proper divisor of n larger than $n/3$ is $n/2$, if it happens to be an integer; this implies that n is even and that the larger of $|C_1|$ and $|C_2|$ is $n/2$. Supposing without loss of generality that $|C_1| = n/2$, we find that $|C_2| = n/2 - 1$, so $n/2 - 1$ also divides n. Since $\gcd(n/2, n/2 - 1) = 1$, n is divisible by their product $n(n-2)/4$, so in particular is at least large as this product. Thus

$$n \geq \frac{n(n-2)}{4} \iff 4 \geq n - 2 \iff n \leq 6,$$

and we can check that $n = 4$ and $n = 6$ both are divisible by $n/2 - 1$. However, $n = 4$ cannot have three conjugacy classes because all groups of order four are abelian (either $\mathbb{Z}/4\mathbb{Z}$ or $(\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$), hence have four conjugacy classes. Meanwhile, the only non-abelian group of order 6 is S_3 (easy to check via Sylow theorems), which indeed has three conjugacy classes

$$\{(1)\}, \quad \{(1\ 2), (1\ 3), (2\ 3)\}, \quad \{(1\ 2\ 3), (1\ 3\ 2)\}.$$

Thus, any finite group with three conjugacy classes is isomorphic to either $\boxed{\mathbb{Z}/3\mathbb{Z}}$ or $\boxed{S_3}$. □

Problem 7. Let K be a field of characteristic p and $f(x) = x^p - x - a$ for some $a \in K$. Show that f is either irreducible in $K[x]$ or decomposes into a product of linear factors.

Solution. The main idea is the identity

$$f(r+k) = (r+k)^p - (r+k) - a = r^k - r - a + k^p - k = r^k - r - a = f(r) \qquad (*)$$

for any $k \in \mathbb{F}^p$ due to Fermat's little theore (strictly speaking, k is in the natural embedding of $\mathbb{F}_p$ in K), so r is a root if and only if $r+k$ is a root.

Let L/K be an extension in which f splits and let α be one of its roots. and let $m(x) \in K[x]$ be the minimal polynomial of α in K, so the minimal polynomial of $\alpha + k$ is $m(x-k)$. If R_k is the set of roots of $m(x-k)$ for $k \in \mathbb{F}_p$, then the union of the R_k is $\{\alpha + k : k \in \mathbb{F}_p\}$. Moreover if two of the R_k intersect then they are the same, by the minimality of m (else take the gcd of the corresponding $m(x-k)$s to get a lower degree minimal polynomial for one of the $\alpha + k$). In particular the distinct sets among the R_ks form a partition of the p-element set of roots of f in L into sets of equal size, so the R_ks all either have size 1 or size p, corresponding respectively to splitting into linear factors or being irreducible in K. □

Problem 8. Consider the set $\mathbb{Q}$ of rational numbers as a group under addition. Prove that there is no proper subgroup $G \leq \mathbb{Q}$ of finite index.

Solution. Suppose $G \leq \mathbb{Q}$ has finite index n and let $\pi \colon \mathbb{Q} \to \mathbb{Q}/G$ be the projection homomorphism. Let $x \in \mathbb{Q}$ be arbitrary. By Lagrange's theorem, $\pi(x/n)$ has order dividing n in $\mathbb{Q}/G$, so $\pi(x) = n\pi(x/n)$ is the identity in $\mathbb{Q}/G$. Thus $x \in \ker \pi = G$, with the equality following from the first isomorphism theorem, implying G contains all the rational numbers. □

Problem 9. Let p be a prime number. The symmetric group S_p acts in a natural way on the set $X = \{1, 2, \ldots, p\}$. Prove that the induced action of a subgroup $G \leq S_p$ is transitive if and only if G contains a p-cycle.

Solution. If G contains a p-cycle then let σ be the cycle; for any $i, j \in X$, the set $\{i, \sigma(i), \ldots, \sigma^{p-1}(i)\}$ is equal to X because it has p distinct elements (since the order of σ is p), hence some power of σ yields j when applied to i. This means the element of S_p corresponding to this power of σ sends i to j, i.e. the action is transitive.

If the induced action of G is transitive, then the action of S_p only has one orbit of size p. By the orbit-stabilizer theorem, p divides the order of G, so by Cauchy's theorem G contains an element of order p. The only elements of order p in S_p are the p-cycles, so we are done. □

Problem 10. Classify, up to isomorphism, all groups of order 45.

Solution. Let G be a group of order 45. By the Sylow theorems, since $45 = 3^2 \cdot 5$, the number of subgroups of order 5 is a divisor of 9 and is 1 mod 5, which means that G has one subgroup P_5 of order 5. Therefore, for any g, the subgroup gP_5g^{-1} also has order 5, so $gP_5g^{-1} = P_5$, i.e. P_5 is normal in G. Similarly, the number of subgroups of order 9 is a divisor of 5 and is 1 mod 3, so G has one subgroup P_9 of order 9, which by the same logic is normal in G.

We can then see that $P_5P_9 = \{ab : a \in P_5, b \in P_9\}$ is a subgroup of G because P_5 and P_9 are normal in G, and it has order divisible by both 5 and 9 because P_5, P_9 are subgroups of P_5P_9. Thus, P_5P_9 is a subgroup of G with order at least 45, which means that it is equal to G. Furthermore $P_5 \cap P_9 = \{1\}$ because if P_9 contained some element $g \in P_5$ not equal to 1 then, since P_5 is isomorphic to $\mathbb{Z}/5\mathbb{Z}$, g generates P_5 so P_9 contains P_5, which is nonsensical because 5 doesn't divide 9. To finish we use the following lemma:

Lemma 10.1. Suppose N and K are normal in G such that $NK = G$ and $N \cap K = \{1\}$. Then elements of N commute with elements of K, and in particular $G \cong N \times K$.

By this lemma, we find that $G \cong P_5 \times P_9$. As mentioned above, $P_5 \cong \mathbb{Z}/5\mathbb{Z}$. For P_9, there are only two cases: either there is an element of order 9, in which case $P_9 \cong \mathbb{Z}/5\mathbb{Z}$ and $\boxed{G \cong \mathbb{Z}/45\mathbb{Z}}$, or every non-identity element of P_9 has order 3. In the latter case, the class equation implies that 3 divides the order of the center $|Z(P_5)|$ (because all non-fixed points have a conjugacy class consisting of 3 elements). If $Z(P_5)$ has only three elements, then suppose g isn't in $Z(P_5)$; it clearly commutes with itself and every element of $Z(P_5)$, so the centralizer of P_5 has order at least 4, meaning it has order 9, but this contradicts $g \notin Z(P_5)$. Therefore $Z(P_5) = P_5$, i.e. P_5 is abelian. The only possibility, then, is $P_5 \cong (\mathbb{Z}/3\mathbb{Z}) \times (\mathbb{Z}/3\mathbb{Z})$, so that $\boxed{G \cong (\mathbb{Z}/15\mathbb{Z}) \times (\mathbb{Z}/3\mathbb{Z})}$. □

Problem 11. List, up to isomorphism, all $\mathbb{F}_2[x]$-modules with exactly 8 elements.

Solution. Let M be an $\mathbb{F}_2[x]$-module with 8 elements. Since M is finite and $\mathbb{F}_2[x]$ isn't, M must be a torsion module, so it is isomorphic to a direct sum of modules of the form $\mathbb{F}_2[x]/(p(x)^k)$ for some irreducible polynomial p by the characterization of finitely generated modules over PIDs. Noting that $\mathbb{F}_2[x]/(f(x))$ has $2^{\deg f}$ elements because the powers of x in the quotient give an $\mathbb{F}_2$-basis of size $\deg f$, the possible decompositions of M are as follows:

(1) $\mathbb{F}_2[x]/(x) \oplus \mathbb{F}_2[x]/(x) \oplus \mathbb{F}_2[x]/(x)$

(2) $\mathbb{F}_2[x]/(x) \oplus \mathbb{F}_2[x]/(x) \oplus \mathbb{F}_2[x]/(x+1)$

(3) $\mathbb{F}_2[x]/(x) \oplus \mathbb{F}_2[x]/(x+1) \oplus \mathbb{F}_2[x]/(x+1)$

(4) $\mathbb{F}_2[x]/(x+1) \oplus \mathbb{F}_2[x]/(x+1) \oplus \mathbb{F}_2[x]/(x+1)$

(5) $\mathbb{F}_2[x]/(x^2) \oplus \mathbb{F}_2[x]/(x)$

(6) $\mathbb{F}_2[x]/(x^2) \oplus \mathbb{F}_2[x]/(x+1)$

(7) $\mathbb{F}_2[x]/((x+1)^2) \oplus \mathbb{F}_2[x]/(x)$

(8) $\mathbb{F}_2[x]/((x+1)^2) \oplus \mathbb{F}_2[x]/(x+1)$

(9) $\mathbb{F}_2[x]/(x^2+x+1) \oplus \mathbb{F}_2[x]/(x)$

(10) $\mathbb{F}_2[x]/(x^2+x+1) \oplus \mathbb{F}_2[x]/(x+1)$

(11) $\mathbb{F}_2[x]/(x^3+x+1)$

(12) $\mathbb{F}_2[x]/(x^3+x^2+1)$

(13) $\mathbb{F}_2[x]/(x^3)$

(14) $\mathbb{F}_2[x]/((x+1)^3)$

This list contains all possible decompositions of M because the irreducible polynomials in $\mathbb{F}_2[x]$ of degree 1 are x, $x+1$; the only irreducible polynomial in $\mathbb{F}_2[x]$ of degree 2 is x^2+x+1; and the irreducible polynomials in $\mathbb{F}_2[x]$ of degree 3 are x^3+x+1, x^3+x^2+1 (we can verify the latter two fairly quickly by just listing out all the polynomials of degree 2, 3 and checking for linear factors; there are only six polynomials to check because the leading and constant coefficients must be 1). Furthermore, any two decompositions in the list are nonisomorphic by the structure theorem again, so the list enumerates all the isomorphism classes of 8-elements $\mathbb{F}_2[x]$-modules. □